EE3001: Solid State Devices
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Assignment #5

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed in class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following values by default unless otherwise specified: Si is the default semiconductor; SiO_2 is the default oxide; T @room temperature is 300 K; n_i@room temp. = $10^{10} cm^{-3}$; k_BT=0.026 eV (at room temperature); $E_{G,Si}$ = 1.12 eV and temperature independent; at 300 K, N_c=$3.2 \times 10^{19} cm^{-3}$, N_v=$1.8 \times 10^{19} cm^{-3}$, $\mu_n = 1350 cm^2/V.s$, $\mu_p = 450 cm^2/V.s$; Permittivity for vacuum: ϵ_0=8.85×10^{-14}F/cm, relative permittivity: $\epsilon_{r,Si}$=11.8, ϵ_{r,SiO_2}=3.9; pn diode dopants are completely ionized.

1. (2 points) Acceptor and donor concentration in the p- and n-side of an abrupt pn-junction are $N_A = 4 \times 10^{18}\ /cm^3$ and $N_D = 4 \times 10^{17}\ /cm^3$, respectively.

   (a) Sketch (in scale) the electric field distribution from $-X_{p0}$ to $+X_{n0}$ considering the metallurgical junction at the origin. Here X_{p0} and X_{n0} are the penetration depth of the space charge region in p and n regions, respectively.

   (b) Now if p-side doping is modified to $N_A = 8 \times 10^{18}\ /cm^3$, sketch the electric field profile on the same plot.

   (c) How will the plot in (a) appear at an elevated temperature?

2. (2 points) The built-in potential of an abrupt p-n junction is 0.75 V. If the depletion layer width (W) and junction capacitance (C_J) are 1 μm and 10 pF when the reverse bias (V_r) is 1.25 V, what are the values of (a) C_J and (b) W when V_r is 31.25 V?

3. (2 points) At T=300 K, measured voltage dependent inverse squared junction capacitance ($1/C_j^2$) of an abrupt Si p^{++}n junction with cross sectional area of 100 μm^2 predicts an x-intercept of 0.725 V and y-intercept of $10^{30}\ F^{-2}$. If the n-sided doping is given by $M \times 10^{14}\ cm^{-3}$, find the value of M.

4. (2 points) A pn diode has N_A=$10^{16}\ cm^{-3}$ and N_D=$10^{15}\ cm^{-3}$. Another identically doped pn diode has an additional intrinsic layer introduced in between the p- and n-sides leading to a p-i-n structure. The intrinsic or i-layer in the p-i-n structure has a length of 2 μm.

   (a) Under zero bias, calculate the ratio of the depletion capacitance of the p-i-n ($C_{j,pin}(0)$) and p-n ($C_{j,pn}(0)$) diodes.

   (b) Under -5 V reverse bias, calculate the depletion capacitances of the p-i-n ($C_{j,pin}(-5)$) and p-n ($C_{j,pn}(-5)$) diodes.

(c) Find the ratios $(C_{j,pn}(-5)/C_{j,pn}(0))$ and $(C_{j,pin}(-5)/C_{j,pin}(0))$.

(d) Name an application where such p-i-n structure can be useful.

5. (2 points) A p^+n diode has non-uniform n-sided doping given by $N_D(x)$=$N_{D0}(x/x_0)^m$ cm^{-3}. Such a diode is being used as a bias-dependent variable capacitor ($C(V)$) in a local oscillator circuit producing a variable resonant frequency $f_r = 1/2\pi\sqrt{LC(V)}$. It is found that $f_r \propto V^n$.

   (a) Should the diode be used in forward bias or reverse bias?

   (b) If $m = 0$, find the value of n.

   (c) If $m = 1$, find the value of n.

   (d) What should be the value of m in order to ensure $n = 1$?